# Secure Network Design on Van615 Hotel

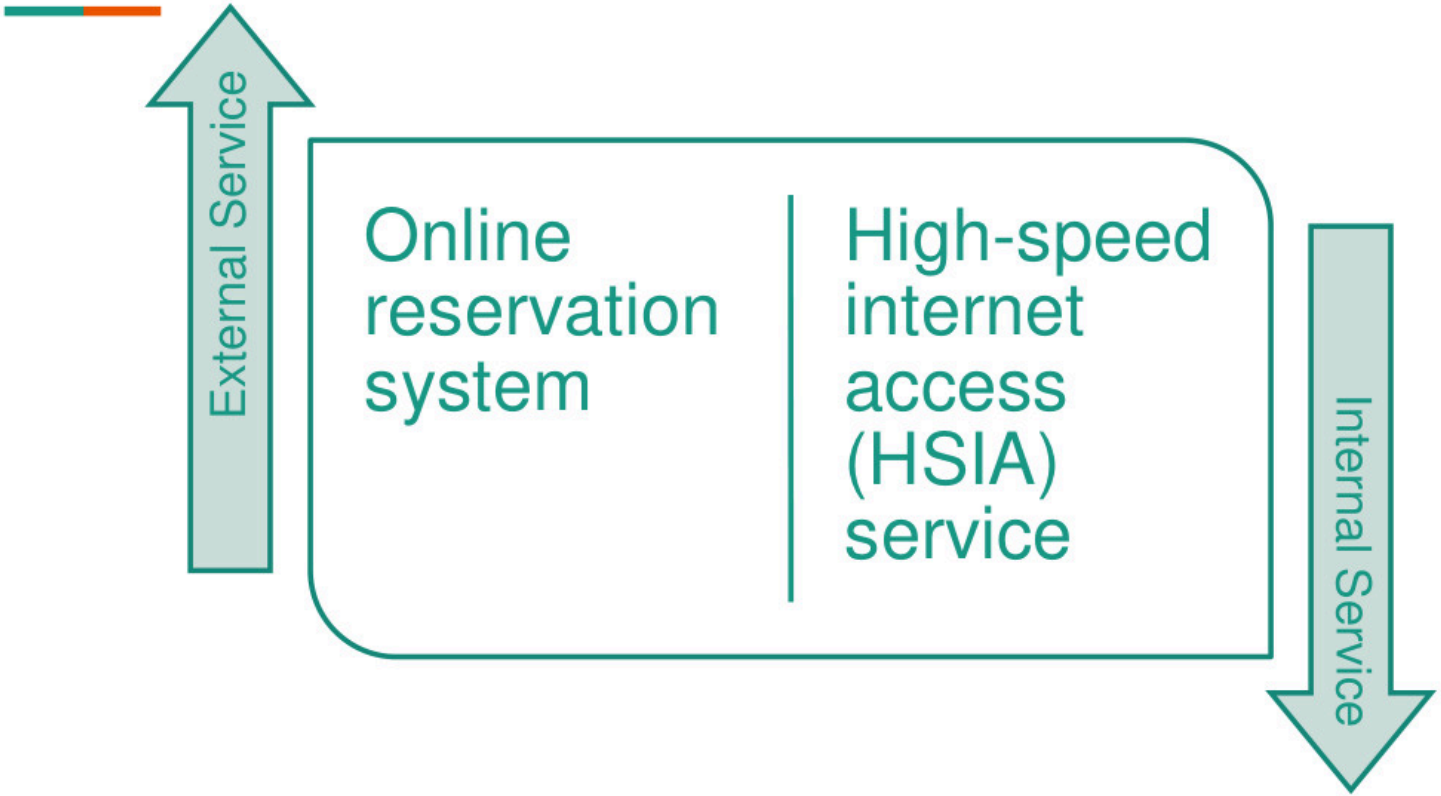


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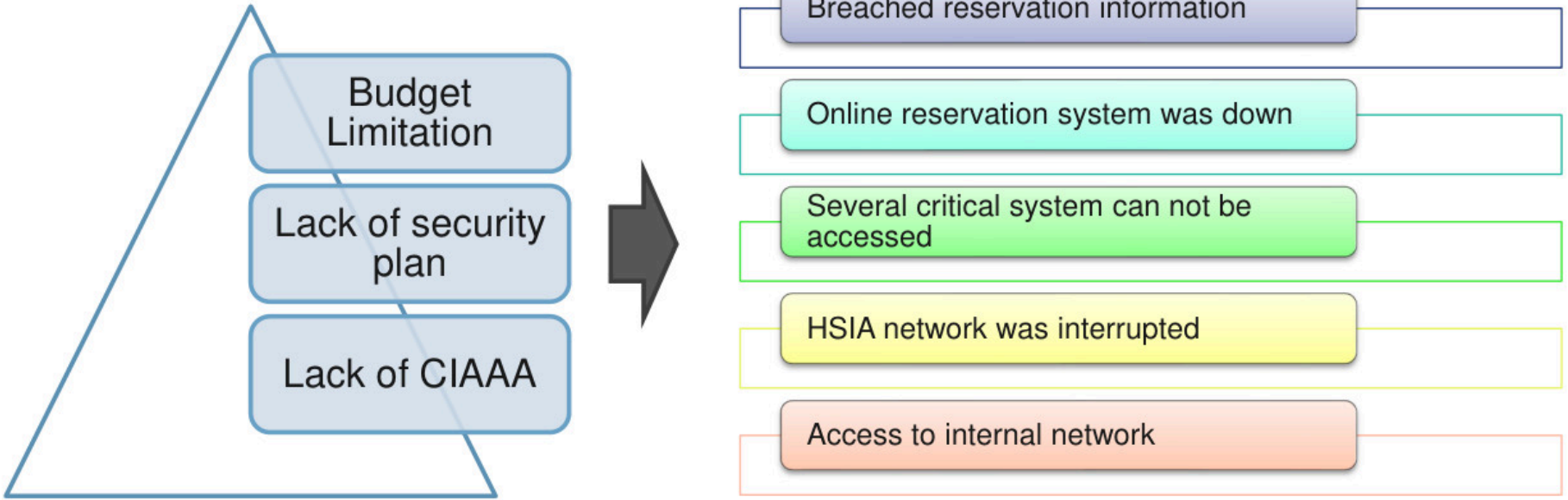
# Outline

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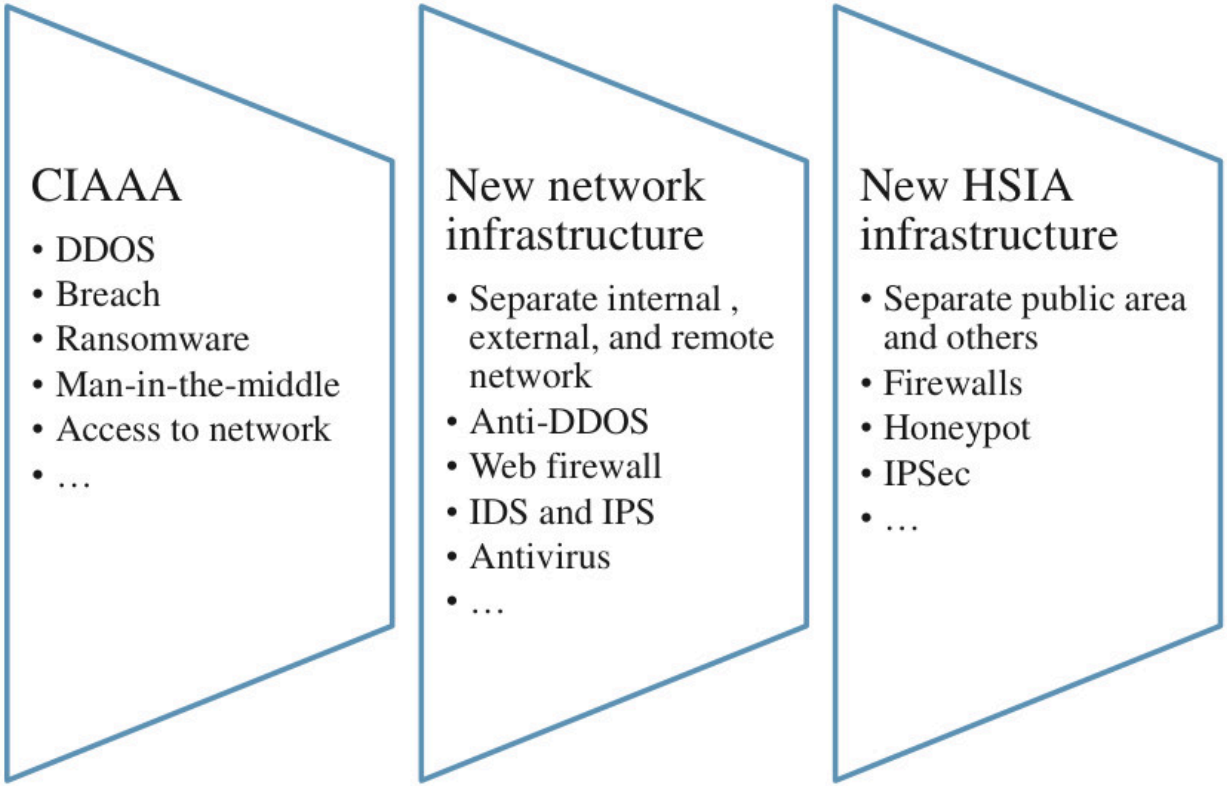
# Introduction



# Introduction



# Goal


CIAAA
• DDOS
• Breach
• Ransomware
• Man-in-the-middle
• Access to network
• …
New network infrastructure
• Separate internal , external, and remote network
• Anti-DDOS
• Web firewall
• IDS and IPS
• Antivirus
• …
New HSIA infrastructure
• Separate public area and others
• Firewalls
• Honeypot
• IPSec
• …

# Implementation of secure network design

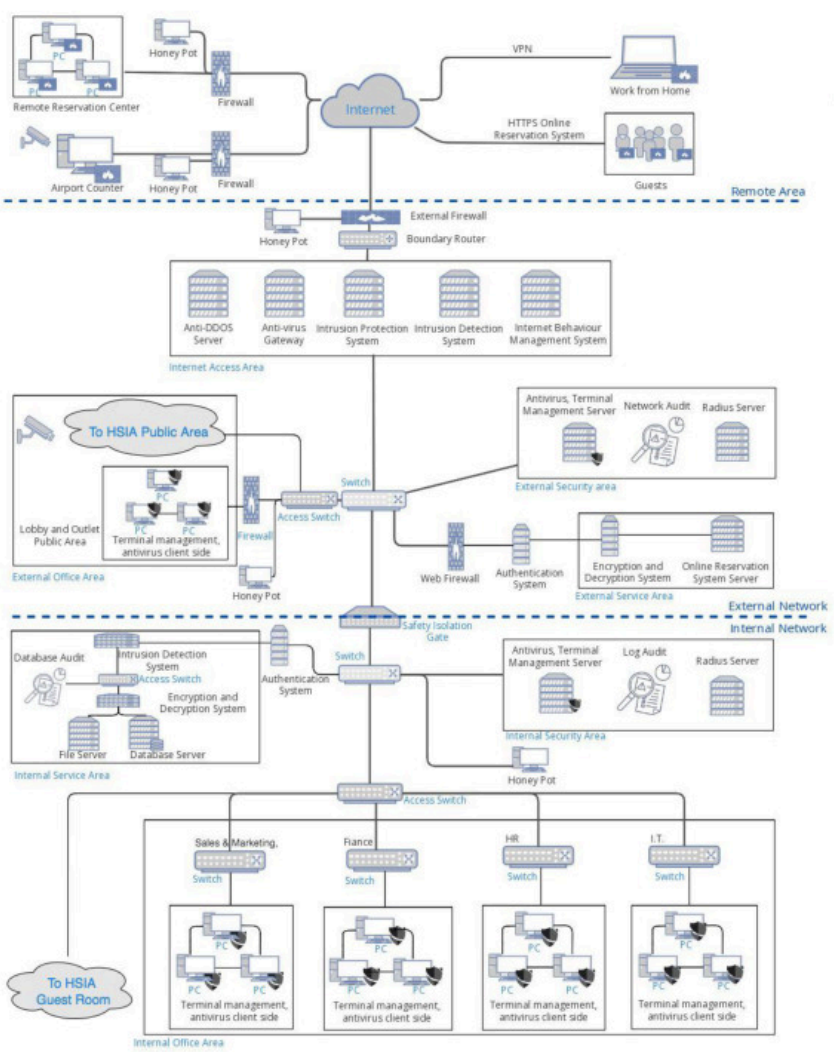
Current incidents could be solved by:

| | |
|---|---|
| Breached reservation information | • Deploy audit systems (active directory) to databases<br>• use a database firewall to stop SQL injection and override operations<br>• Deploy an encryption system for sensitive data |
| Online reservation system was down | • Use firewall to filter the malicious address and deny their service<br>• Set exchangers to identify the fake address and filter the true client<br>• Set anti-DDOS-attack program or other human-checking program |
| Several critical system can not be accessed | • Bayesian Network-based Analysis; Process monitoring<br>• Stack Activity Analyzer, I/O log Analyzer; backup and recovery layers<br>• Honeypots and IDS to screen network logs |
| HSIA network was interrupted | • Use WPA3 and SSL/TLS for encryption<br>• Use VPN for sensitive information<br>• Be wary of fake WiFi networks |
| Access to internal network | • Set up the Radius Server and IP addresses in the hotel's network<br>• Make configuration of 802.1x protocol on the switches<br>• Authorize desktop etc from guests using 802.1x |

# New Hotel network infrastructure

-Remote and external network

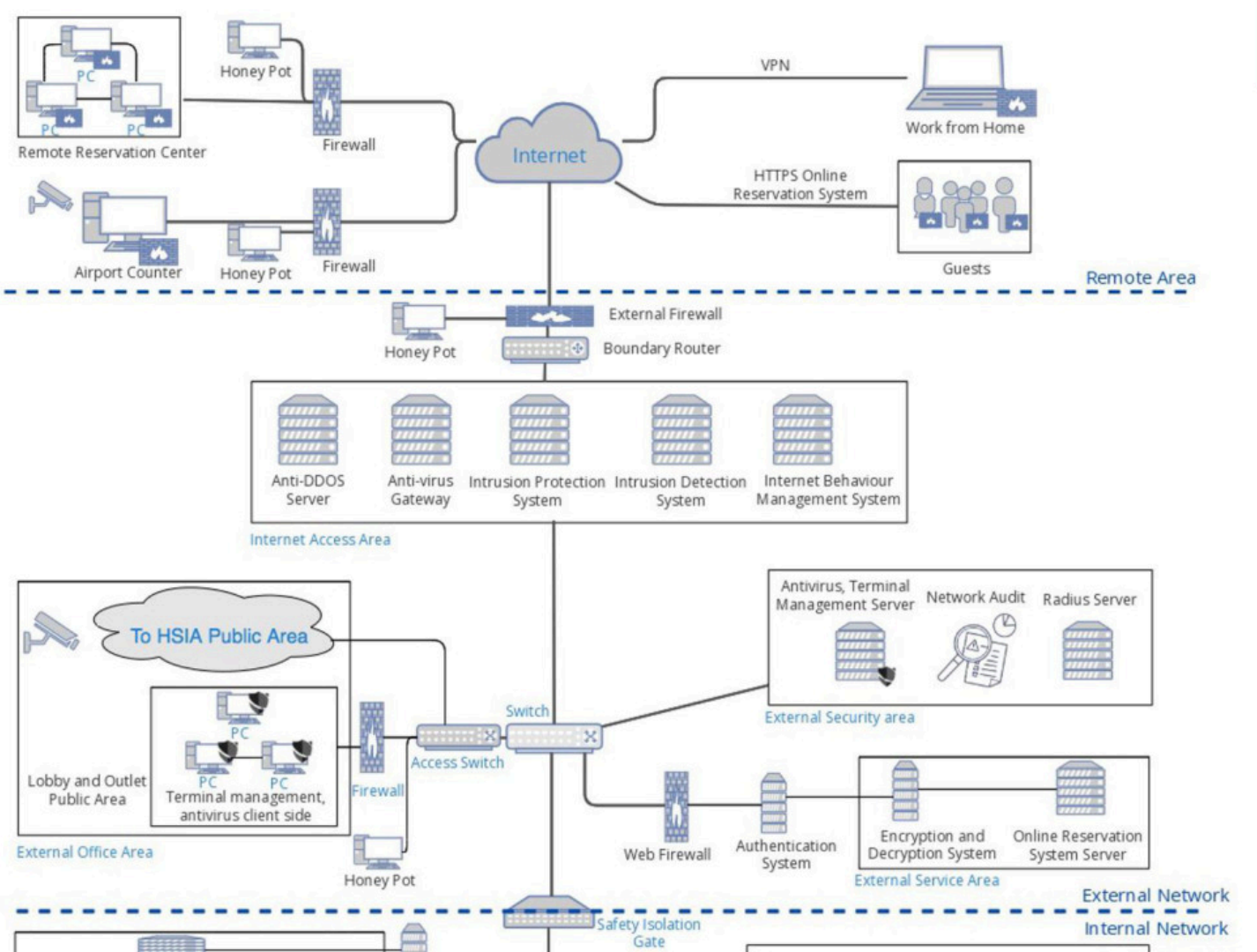
-Internal network



**New Hotel network infrastructure**

-Remote and external network

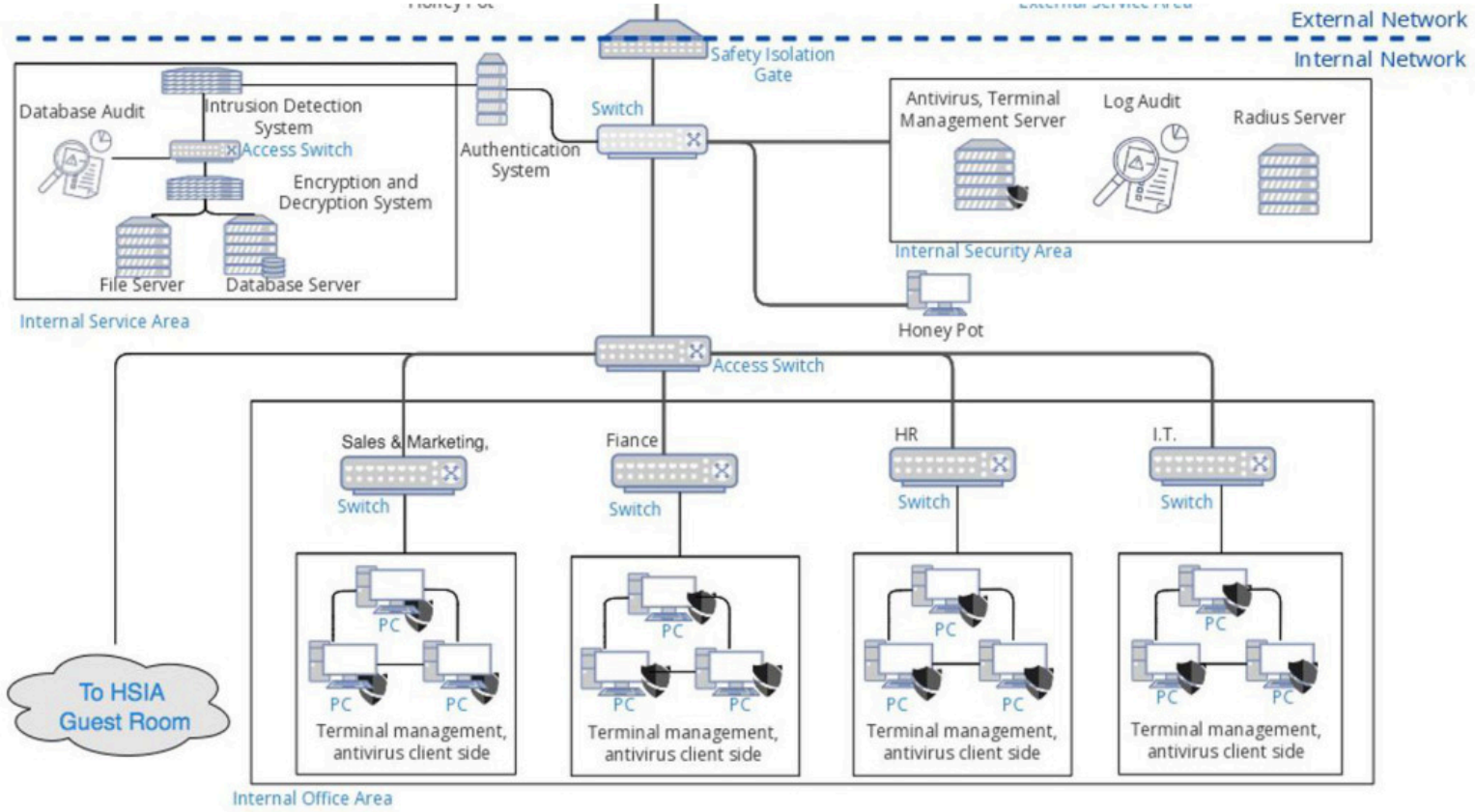
-Internal network



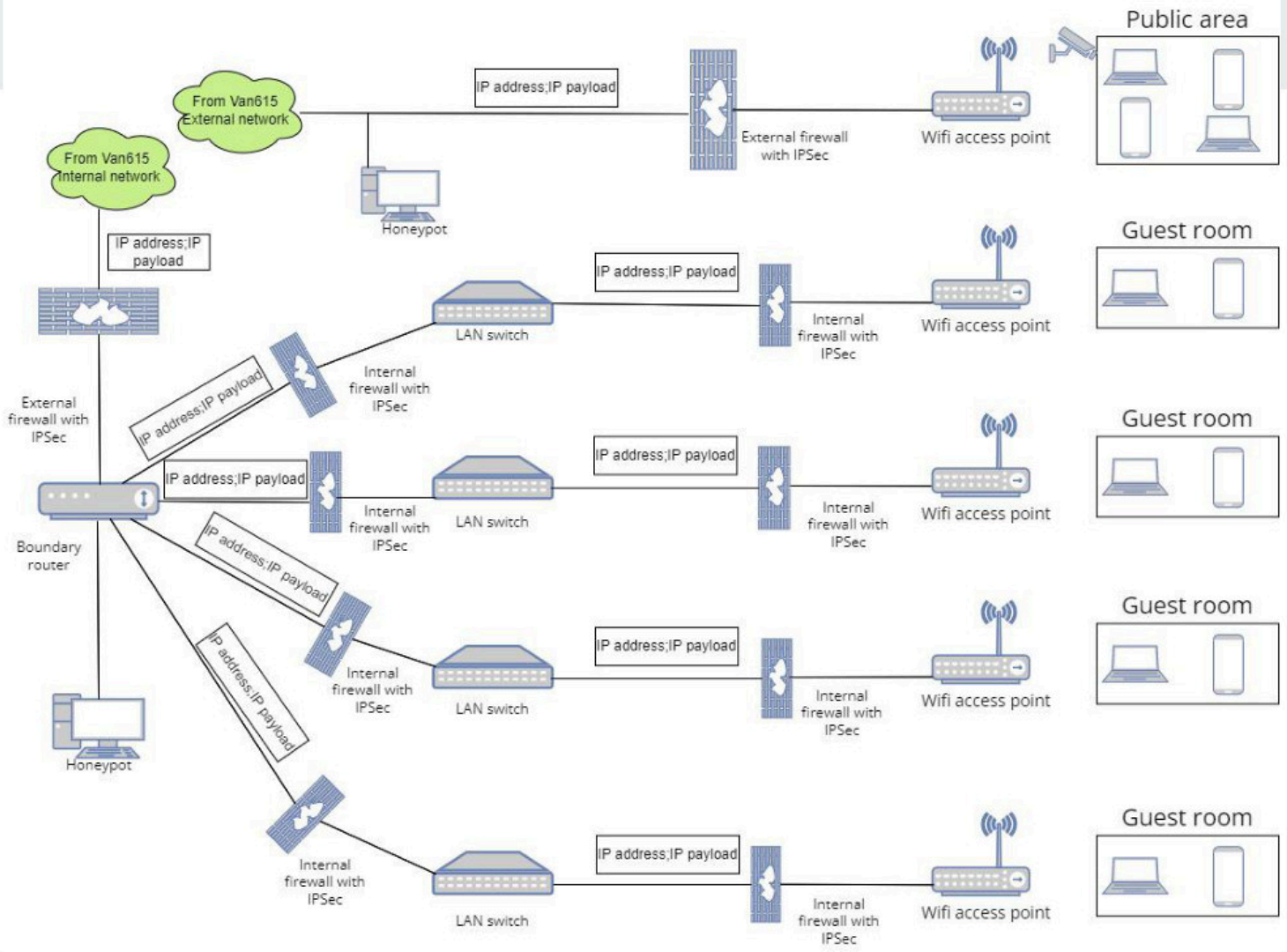
**New Hotel network infrastructure**

-Remote and external network

-Internal network



New HSIA infrastructure based on new Hotel network



# Implementation of secure network design

- Access control matrix

| | Storage &Exec Location | S&M | Finance | HR | IT |
|---|---|---|---|---|---|
| **Customer's Info** | Data Tier | read & write | read & write | n/a | read |
| **Employee's Info** | Data Tier | n/a | read & write | read | read |
| **Payment Info (History payment record)** | Data & Business Tier | read | read& write | n/a | read |
| **Reservation Info and Hotel room availability Record** | Data & Web Tier | read & write | n/a | n/a | read & write |

# Solution and recommendation

- New Van615 Hotel infrastructure
  - Strongly recommend setting Firewalls to reduce threats from the public
  - Highly suggest using Radius for user authentication
  - Strongly recommend setting anti-virus software on the office hosts
  - Deploy honeypots next to some important part (optional)
  - Set IDS and IPS to detect and analyze unusual network behavior and content (optional)
  - Use auditing systems (active directory) (optional)

# Solution and recommendation

- New HSIA infrastructure

Strongly suggest separating the public area and guest room into two different routers using firewalls

Highly recommend the hotel uses firewalls from the Internet

Set internal firewall (optional), it can be attached with the IPSec (optional)

Use the honeypot to the HSIA (optional)

# Solution and recommendation

Other recommendations

- Preparation (update/patch their systems)
- Data storage and backup in another place
- Employee training

# Conclusion

- Analyze the current problems
- Budget and resources are being considered
- Design the structure of this security plan of the Hotel
  - New Hotel network infrastructure
  - New HSIA infrastructure
- Separate internal network and external network
- Recommendations under limited budgets
- Provide an access matrix
- The security measures need constant updates

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# Thank You

## Q&A